Centralni limitni izrek

$X_1, X_2, \ldots$ neodvisne enako porazdeljene (NEP) slučajne spremenljivke z $E(X_1^2) < \infty$.

Označimo $S_n = X_1 + \ldots + X_n$

Za n dovolj velik je

$$P(a \leq S_n \leq b) \approx \Phi\left(\frac{b - nE(X_1)}{\sqrt{n \operatorname{var}(X_1)}}\right) - \Phi\left(\frac{a - nE(X_1)}{\sqrt{n \operatorname{var}(X_1)}}\right)$$

$$\Phi(x) = P(Z \leq x) \; ; \; Z \sim N(0,1)$$

Kocko vržemo 105-krat

a) Oceni vrjetnost da pade < 350 pik.

X_i... pike na i-ti kocki

X_i-ji so enako porazdeljeni: $E(X_i) = 3{,}5$

$P(-\infty \le S_{105} \le 350) \approx$

$$E(X_i^2) = \sum_{i=0}^{6} \frac{1}{6} i^2 = \frac{1}{6} \cdot \frac{6 \cdot 7 \cdot 13}{6} = \frac{7 \cdot 13}{6^2}$$

$$\mathrm{var}(X_i) = -\frac{7^2}{4} + \frac{7 \cdot 13}{6^2} = \frac{35}{12}$$

$$P(S < 350) \approx \Phi\left(\frac{350 - 105 \cdot \frac{7}{2}}{\sqrt{105 \cdot \frac{35}{12}}} \right) = \Phi(-1) \approx 0{,}159\ldots$$

pri aproksimaciji je oboje enako

$$P(S \le 349) = P(S < 350) = 0{,}145$$

Dejanska vrjetnost: $0{,}152$

$P(S < 349{,}5) = \ldots \approx 0{,}152$

V povprečju je srednja vrednost najboljša aproksimacija

b) Oceni vjetnost da 6 pade največ 10-krat

I_i ... na i-tem mestu pade 6

$P(I_i = 1) = \frac{1}{6}$ $\quad I_i \sim Ber(\frac{1}{6})$

$var(I_i) = \frac{1}{6} \cdot \frac{5}{6} = \frac{5}{6^2}$

$$P(S_{105} \leq 10) = P(S < 10{,}5) \approx \Phi\left(\frac{10 - \frac{105}{6}}{\sqrt{105 \cdot \frac{5}{36}}}\right) \approx 0{,}03$$

dejansko: 0,027

visoka relativna napaka

Problem: nesimetrična vjetnost: $\frac{1}{6}$ proti $\frac{5}{6}$

ali pa $\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \ldots$

V sloveniji živi 51% žensk. Koliko klicev mora opraviti v Mediani, da bo v anketi sodelovalo vsaj pol žensk z vrjetnostjo vsaj 0,9?

X_i ... poklicana je ženska indikator

$X_i \sim Ber(p) \qquad p = 0{,}51$

$X_1 + \ldots + X_n = S_n$

$P(S_n \geq \frac{n}{2}) \geq 0{,}9$

$P(S_n \leq \frac{n}{2}) < 0{,}1$

$P(S_n \leq \frac{n}{2}) \approx \Phi\left(\overbrace{\frac{\frac{n}{2} - np}{\sqrt{np(1-p)}}}^{z}\right) < 0{,}1$

$\Phi(z) < 0{,}1$

$z < -1{,}29$

$\sqrt{n}\left(\frac{1}{2} - p\right) < -1{,}29 \sqrt{p(1-p)}$

$\sqrt{n} > \frac{1{,}29 \sqrt{p(1-p)}}{-(\frac{1}{2} - p)}$

n mora biti vsaj 4000

mora biti vsaj

3. $X_i \sim \text{Gama}(a, \lambda)$ neod

Izračunaj $P(X_1 + \dots + X_n \leq \frac{cna}{\lambda})$

Momentna rodovna funkcija $M_X(t) = E(e^{tX})$

X, Y neodvisni:

$$M_{X+Y}(t) = E(e^{t(X+Y)}) = E(e^{tX} \cdot e^{tY}) \overset{\text{neodvisnost}}{=} E(e^{tX})E(e^{tY}) =$$
$$= M_X(t) \cdot M_Y(t)$$

$$M_{X_1}(t) = E(e^{tX_1}) = \int_x e^{tx} \frac{\lambda^a x^{a-1}}{\Gamma(a)} e^{-\lambda x} \, dx =$$

$$\frac{\lambda^a}{\Gamma(a)} \int_{-\infty}^{\infty} e^{x(t-\lambda)} x^{a-1} \, dx \dots = \frac{1}{(1-\frac{t}{\lambda})^a}$$

Momentna rodovna funkcija enolično določa porazdelitev

$$M_{X_1+X_2}(t) = \frac{1}{(1-\frac{t}{\lambda})^{2a}} = M_{\Gamma(2a,\lambda)}(t)$$

$$\Rightarrow \sim \Gamma(2a, \lambda) \quad \Longrightarrow S_n \sim \Gamma(na, \lambda)$$

$$P(S_n \leq \frac{cna}{\lambda}) = \int_{-\infty}^{\frac{cna}{\lambda}} \frac{\lambda^{na} x^{na-1} e^{-\lambda x}}{\Gamma(a)} \, dx =$$

$$= \int_0^{\frac{cna}{\lambda}} \frac{(\lambda x)^{na-1} \lambda e^{-\lambda x}}{\Gamma(a)} \, dx = \int_0^{cna} \frac{1}{\Gamma(a)} (u)^{na-1} e^{-\lambda u} \, du$$

DN: uporabi centralni limitni izrek

Ocene za napake pri CLI

če imajo X_i tudi tretji moment

velja ocena

$$\sup_{x\in\mathbb{R}} \left|P(S_n \leq x) - \Phi\left(\frac{x - mE(X_1)}{\sqrt{n\,\mathrm{var}(X_1)}}\right)\right| \leq \frac{0{,}5583}{(n\,\mathrm{var}(X_1))^{\frac{3}{2}}}\, nE(|X_1 - E(X_1)|^3)$$

da se posplošiti za primer

ko $X_1, \ldots, X_n$ niso enako porazdeljeni:

6.3

① $X_1, X_2, \ldots$ neodv. in $X_k \sim \exp(\frac{1}{k})$

Ali v tem primeru velja CLI?

$$S_n = \sum_{i=1}^{n} X_i \qquad E(X_k) = k \qquad var(X_k) = k^2$$

$$var(S_n) \underset{\text{neodvisnost}}{=} \sum_{i=1}^{n} var(X_i) = \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$E(|X_k - E(X_k)|^3)$$

$$\gamma_1^3 = E(|X_1 - E(X_1)|^3) = \frac{12}{e} - 2$$ ← namig

$$X_k \overset{d}{=} k X_1$$

$$P(X_k > x) = e^{-\frac{1}{k}x} \qquad P(kX_1 > x) = P(X_1 > \frac{x}{k}) = e^{-\frac{1}{k}x}$$

↑ preživetvena funkcija

$$\Rightarrow E(|X_k - E(X_k)|^3) = E(|kX_1 - kE(X_1)|^3) = k \cdot \gamma_1^3$$

Oglejmo si

$$\frac{\sum_{k=1}^{n} E(|X_k - E(X_k)|^3)}{var(S_n)^{\frac{3}{2}}} = \frac{\gamma^3 (1 + \ldots + n^3)}{\left(\frac{n(n+2)(2n+1)}{6}\right)^{\frac{3}{2}}} = \frac{\gamma^3 \frac{n^2(n+1)^2}{4}}{\left(\frac{n(n+2)(2n+1)}{6}\right)^{\frac{3}{2}}}$$

$$= O\left(\frac{n^4}{n^{\frac{9}{2}}}\right) = O\left(\frac{1}{\sqrt{n}}\right) \longrightarrow 0$$

$\Rightarrow$ CLI velja

②

$p \in (0,1)$ $\quad S \sim Bin(n,p)$

Oceni: $\left| P(S \leq x) - \Phi\left(\frac{x-np}{\sqrt{np(1-p)}}\right)\right| = *$

Obstajajo $X_1, \dots, X_n \sim Bernuli(p)$ neodvisne

$\quad S = \sum X_i$

Poglejmo si

$$E(|X_1 - \underbrace{E(X_1)}_{p}|^3) = E(|X_1 - p|^3) =$$

$$= |1-p|^3 \cdot p + (1-p)|p^3| = p(1-p)((1-p)^2 + p^2) \leq p(1-p)$$

$$* \leq \frac{c \; np(1-p)}{(np(1-p))^{\frac{3}{2}}} = \frac{c}{\sqrt{np(1-p)}}$$

(3)

$X_1, \ldots, X_{100}$ neodv. enako porazdeljene

$X_1 \sim \begin{pmatrix} -1 & 1 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$

$$S = 10X_1 + X_2 + \ldots + X_{100}$$

Oceni vrjetnost

$P(S<41)$

$$\star = P(S<41 \mid X_1=-1)\overbrace{P(X_1=-1)}^{\frac{1}{3}} + P(S<41 \mid X_1=0)\overbrace{P(X_1=0)}^{\frac{2}{3}}$$

$$P(S<41 \mid X_1=-1) = P(\tilde{S}<51)$$

$$P(S<41 \mid X_1=1) = P(\tilde{S}<31)$$

$$\star \approx \frac{1}{3}\Phi\left(\frac{50-99\cdot\frac{1}{3}}{\sqrt{99\cdot\frac{8}{9}}}\right) + \frac{2}{3}\Phi\left(\frac{30-33}{\sqrt{88}}\right) \approx 0{,}57\ldots$$

Točen rezultat $\approx 0{,}569$

če bi uporabili CLI direktno na S bi dobili $\approx 0{,}6$

$X_1, X_2, \ldots$ neodv $\quad X_n \sim \begin{pmatrix} -n & 0 & n \\ \frac{1}{2n^2} & 1-\frac{1}{n^2} & \frac{1}{2n^2} \end{pmatrix}$

Pokaži, da CLI ne velja

Recimo da velja CLI

Potem $\lim_{n\to\infty} P(X_1+\ldots+X_n$

$$E(S_n) = \sum E(X_n) = 0$$

$$\mathrm{var}(X_n) = E(X_n^2) = 1 \implies \mathrm{var}(S_n) = n$$

$$W_n = \frac{S_n - 0}{\sqrt{n}} \qquad P(W_n \in [-\tfrac{1}{2}, \tfrac{1}{2}]) = \Phi(\tfrac{1}{2}) - \Phi(-\tfrac{1}{2}) =$$

$$= 1-2\,\Phi(-\tfrac{1}{2}) \approx 0{,}38$$

$n \geq 4$

$$P(-\tfrac{1}{2} \leq W_n \leq \tfrac{1}{2}) = P(-\tfrac{1}{2} \leq \frac{S_n}{\sqrt{n}} \leq \tfrac{1}{2}) =$$

$$= P(\tfrac{\sqrt{n}}{2} \leq S_n \leq \tfrac{\sqrt{n}}{2}) \geq P(X_2 = X_3 = \ldots = X_n = 0)$$

$\vdots$

$=$

$$\frac{n+1}{2^n} > \frac{1}{2} \quad \text{za } n \geq 4 \quad \times$$

Naj bodo $X_1, X_2, \ldots$ slučajne spremenljivke z vrednostmi sl. spr. $(X_n)_{n\in\mathbb{N}}$ konvergira proti sl. spr. X v porazdelitvi, če

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x) \quad \text{za } \forall x \in \mathbb{R}, \text{ kjer}$$

je F_X zvezna

$X_n = \frac{1}{n}$

Očitno bi želeli, da konvergira proti 0

$$F_{X_n}(x) = \begin{cases} 1 & x \geq \frac{1}{n} \\ 0 & \text{; sicer} \end{cases}$$ ~~Ogle~~

$$F_X(x) = \begin{cases} 1 & x \geq 0 \\ 0 & \text{sicer} \end{cases}$$

Oglejmo si točke nezveznosti

$F_X(0) = 1 \quad F_{X_n}(0) = 0 \quad \forall n \in \mathbb{N}$ ⚡

Oglejmo si, da po definiciji velja povsod kjer je zvezna

Naj bo $x \neq 0$

$F_X(x) = 1$; če $x > 0$

$F_X(x) = 0$ če $x < 0$

$F_{X_n}(x) = 1$ če $x > \frac{1}{n}$

$F_{X_n}(x) = 0$ če $x \leq \frac{1}{n}$

$X_n \xrightarrow{d} X$, kjer $X \sim Pois(\lambda)$

$\lim F_{X_n}(x) = F_X(x)$, kjer je F_X zvezna

$$P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

$$\lim_{n\to\infty} \frac{P(X_n=k)}{P(X_n=k-1)}$$

$$\frac{P(X_n=k)}{P(X_n=k-1)} = \frac{\binom{n}{k} p_n^k (1-p_n)^{n-k}}{\binom{n}{k-1} p_n^{k-1} (1-p_n)^{n-k+1}} =$$

$$= \frac{n-k+1}{k} \frac{p_n}{1-p_n} \qquad n p_n \to \lambda \Rightarrow p_n \to 0$$

$$\lim \cancel{k} = \frac{np_n}{k(1-p_n)} + \frac{p_n(1-k)}{k(1-p_n)} \xrightarrow[n\to\infty]{} \frac{\lambda}{k}$$

$$P(X_n=k) = \frac{P(X_n=k)}{P(X_n=k-1)} \frac{P(X_n=k-1)}{\dots} \dots\dots \frac{P(X_n=1)}{P(X_n=0)} P(X_n=0)$$

$$= \prod_{i=0}^{k} \frac{\lambda}{i} (1-p_n)^n = \lambda (1-p_n)^k \prod_{i=0}^{k} \frac{1}{i} =$$

$$P(X_n=0) = (1-p_n)^n = (1-p_n)^{\frac{np_n}{p_n}} \Rightarrow e^{-\lambda}$$

$$= \frac{\lambda e^{-\lambda}}{k!}$$

$$\lim F_{X_n}(x) = \lim_{n\to\infty} \sum_{k=0}^{\lfloor x \rfloor} P(X_n=k) = \sum \frac{\lambda e^{-\lambda}}{k!} = F_X(x)$$

(M, d) metrični prostor

$\tilde{M} \subseteq M$

$\tilde{x}_1, \tilde{x}_2, \ldots \ldots \tilde{x}$ sl spr z vred. v $\tilde{M}$

1) $x_n \xrightarrow[n\to\infty]{d} \tilde{x}$ v $(\tilde{M}, d)$ ~~z~~ poln $(M, d) \Rightarrow$

2) če $\tilde{M}$ zaprta in X

$X_1, X_2 \dots$ N EP $\quad \geq E(X_1) = 0$
$var(X_1) = \sigma^2 \quad \Rightarrow E(X_1^2) = \sigma^2$
$var(X_1^2) = \tau^2$

a) kam konvergira
$\sqrt{n}\left(\frac{\sum X_i^2}{n} - \sigma^2\right) =$

$\downarrow$
$S \quad E(X^2)$

$$\frac{\sum x_i^2 - n\sigma^2}{\sqrt{n}}$$

$Y_1, Y_2, \dots$ NEP $\quad E(Y_1^2) < \infty$
Potem $S = \sum Y_i \quad \frac{S - E(Y_1)n}{\sqrt{var(Y_1)n}} \xrightarrow{d} N(0,1)$

$Y_i = X_i^2$
$var(X_i^2) = \tau_2$

$$\tau \frac{S - n\sigma^2}{\sqrt{\tau^2 n}} \longrightarrow \tau \cdot N(0,1) = N(0, \tau^2)$$

b) pokaži $\quad \sqrt{n}\,\bar{X}^2 \xrightarrow{d} 0$

$$\bar{X} = \frac{X_1 + \dots X_n}{n} =$$

$\sqrt{n}\,\bar{X} \cdot \bar{X} \qquad \bar{X} \xrightarrow{d} 0 \leftarrow E(X_1)$
Po ~~izrek~~ ~~slutsko~~ šibkem ~~zakon~~ zakonu velikih števil
Pot izrekun slutskega

$\bar{X} \cdot (\sqrt{n}\,\bar{X})$
$$\sqrt{n}\,\bar{X} = \sigma \frac{X_1 + \dots + X_n - 0 \cdot n}{\sqrt{n}\,\sigma} \xrightarrow{d} N(0, \sigma^2)(x)$$

$\rightarrow \sqrt{n}\,\bar{X}^2 \xrightarrow{d} 0$

c) $\quad M_n^2 = \frac{\sum (X_j - \bar{X})^2}{n}$

kam konvergira $\sqrt{n}\,(M_n^2 - \sigma^2)$
$$\frac{\sum (X_j - \bar{X})^2 - n\sigma^2}{\sqrt{n}} =$$
$$= \frac{\sum X_j^2 - 2\sum X_j \bar{X} + \sum \bar{X}^2 - n\sigma^2}{\sqrt{n}}$$

$\longrightarrow N(0, \tau^2)(x) + 0 + 0$

Naj bodo $X_1, X_2 ..$ sluč. sprem. definirane na $(\Omega, \mathcal{F}, P)$. Pravimo da X_n konvergira k sluč. spr. X

a) v vrjetnosti, če $\forall \varepsilon > 0. \lim_{n \to \infty} P(|X_n - X| > \varepsilon) = 0$

b) skoraj gotovo če $P(\lim X_n = X) = 1$

$$P(\{\omega \in \Omega \mid \lim X_n(\omega) = X(\omega)\}) = 1$$

c) $E(f(X_n)) \xrightarrow{d} E(f(X))$

$P \longrightarrow d$

$d \not\longrightarrow P$

$X(0,1)$

$X_n = -X$

$X_n \xrightarrow{d} X$

$\lim P(|X_n - X| > \varepsilon) =$

$\lim (P(2|X| > \varepsilon) =$

$\lim P(|X| > \frac{\varepsilon}{2}) = P(|X| > \frac{\varepsilon}{2})$

$\neq 0$ za $\varepsilon = 1$

$P \Rightarrow d$

$\forall \varepsilon > 0 \lim P(|X_n - X| > \varepsilon) = 0$

$E ~~(R(X_n)~~)$ f zvezna, omejena funkcija

$E(f(X_n))$ $\quad I_n := \{ |X_n - X| > \varepsilon$

$E(f(X_n)) = E(f(X_n)(I_n + I_n^c)) =$

$= E(f(X_n) I_n) + E(f(X_n) I_n^c$

$E(|f(X_n) - f(X)|) = \underbrace{E(|\ldots| \mathbb{1}_{I_n})}_{P(X_n - X) > \varepsilon) = 0} + E(|\ldots| \mathbb{1}_{I_n^c}) < C\varepsilon$ $\quad \overset{C\varepsilon}{\vee}$

[illegible]

$X_n \xrightarrow{s.g} X$ če $P(\lim_{n\to\infty} X_n = X) = 1$

oziroma $P(\{\omega\in\Omega \mid \lim X_n(\omega) = X(\omega)\}) = 1$

$s.g \Rightarrow P$

$P \not\Rightarrow s.g$

$s.g \Rightarrow P$

zraven so lahko omege

$$\{\lim_{n\to\infty} X_n = X\} = \{\forall \varepsilon>0, \exists N>0.\ n\geq N \Rightarrow |X_n - X| < \varepsilon\}$$

$$\bigcap_{\varepsilon>0}\bigcup_{N\geq 0}\bigcap_{n\geq N}\{|X_n-X|<\varepsilon\} =$$

$$= \bigcap_{m\in\mathbb{N}} \underbrace{\bigcup_{N\geq 0}\bigcap_{n\geq N}\{|X_n-X|<\tfrac{1}{m}\}}_{A_m}$$ padajoče v m

$$= \bigcup_{N\geq 0}\underbrace{\bigcap_{n\in\mathbb{N}}\{|X_n - X|<\tfrac{1}{m}\}}_{B_N}$$ naraščajoče v N

Vemo:

$$P\Big(\bigcap_{m\in\mathbb{N}} A_m\Big) = 1$$

Vemo $\lim_{m\to\infty} P(A_m) = 1$

$\|$

$\lim_{m\to\infty} P\Big(\bigcup_{N\geq 0} B_N\Big) = 1$ $\quad A_m \supseteq \bigcap A_m \Rightarrow 1 = P(\bigcap A_m) \leq P(A_m) \leq 1$

$\|$

$\lim_{N\to\infty} P(B_N) = 1$ $\quad B_N \subseteq \{|X_N - X| < \tfrac{1}{m}\}$

$\leq$

$\lim_{N\to\infty} P(\{|X_N - X| < \tfrac{1}{m}\}) = 1$

$P \neq$ s.g.

$X_1(\omega) = \mathbb{1}_{[0,1]}(\omega)$

$X_2(\omega) = \mathbb{1}_{[0,\frac{1}{2}]}$ $\quad X_3 = \mathbb{1}_{[\frac{1}{2}, 1]}$

$X_4 = \mathbb{1}_{[0,\frac{1}{4}]}$ $\quad X_5 = \mathbb{1}_{[\frac{1}{4},\frac{1}{2}]}$, $\quad X_6 = \mathbb{1}_{[\frac{1}{2},\frac{3}{4})}$

$X = 0$

X_n :

Naj bo M in M' metrična prostora in $g: M \to M'$ preslikava s točkami (s množico) nezveznosti $D_g = \{x \in M \mid g \text{ ni zvezna v } x\}$

Naj bodo $X_1, X_2, \dots X$ sluč. spr. z vrednostmi v M in naj bo $P(X \in D_g) = 0$.

če $X_n \xrightarrow[n\to\infty]{d, P, s.g.} X \Rightarrow g(X_n) \xrightarrow[n\to\infty]{d, P, s.g.} g(X)$

27.3

$X_n \xrightarrow{P} 0 \qquad \sqrt{n}\, X_n \xrightarrow{d} N(0,\sigma^2)$

Privzemimo da je $P(X_n=0)=0$

Po izreku gre $\sin(X_n) \xrightarrow{P} \underset{\sin(0)}{\underset{\shortparallel}{0}}$

$$\sqrt{n}\,\sin(X_n) = \frac{X_n}{X_n}\sqrt{n}\,\sin(X_n) \sim 1\cdot N(0,\sigma^2) = N(0,\sigma^2)$$

gre prot 1 je konst

SKLEPNA STATISTIKA

Pri vrjetnosti pri neznani vrjetnostni meri $\mathbb{P}$ in znanem dogodku A računamo $\mathbb{P}(A)$

Pri statistiki opazimo, da se je zgodil dogodek A, ne vemo pod katero verjetnostno mero $\mathbb{P}_\vartheta$; $\vartheta \in \Theta$.

Radi bi sklepali pri katerem $\vartheta \in \Theta$ se dogodek A zgodi z največjo vrjetnostjo

Zgled:

Kovanci z vrjetnostjo grba $\vartheta \in [0,1]$

Pri vrjetnosti fiksiramo in odgovarjamo na vprašanje, da pri prvih petih metih opazimo 1 grb

$X \sim \mathrm{Bin}(5, \vartheta)$

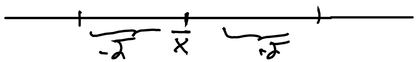
Iščemo ϑ

① $x_1, x_2 \dots x_n$ neodv. enako porazdeljene

$X_1 \sim N(\mu, \sigma^2)$

Recimo da je σ^2 poznan. Poišči interval zaupanja za stopnjo napake α

$$\bar{X} \xrightarrow[n\to\infty]{\text{zakon velikih štev.}} \mu = E(X_1)$$



$$P(\mu \in (\bar{x}-\delta, \bar{x}+\delta)) \geq 1-\alpha$$

$$\parallel$$

$$P(\bar{X}-\delta < \mu < \bar{x}+\delta) \geq 1-\alpha$$

$$\parallel$$

$$P(|\bar{X}-\mu| > \delta) \leq \alpha$$

$$\bar{X} = \frac{X_1 + \dots + X_n}{n} \sim N(\mu, n\sigma^2)$$

$$\mathrm{var}(\bar{X}) = \frac{n\sigma^2}{n^2}$$

$$Z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} = \sqrt{n}\,\frac{\bar{X}-\mu}{\sigma} \sim N(0,1)$$

$$1 - \Phi\left(\frac{\sqrt{n}}{\sigma}\delta\right) = P\left(Z > \frac{\sqrt{n}}{\sigma}\delta\right) \leq \frac{\alpha}{2}$$

$$\delta_c \geq \frac{1{,}96}{\sqrt{n}}$$

② Kot prej μ σ nista poznane

Interval zaupanja za σ^2 ?

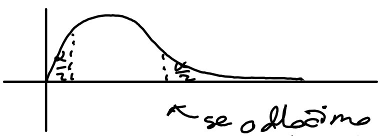
$$\sigma = E((X-\mu)^2)$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)^2 \xrightarrow[n\to\infty]{d} \sigma^2$$

sicer $$\boxed{\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2}$$

izkaže se, da je $\frac{(n-1)\hat{\sigma}^2}{\sigma^2} = Q \sim \chi^2_{n-1}$

Q zadene vrednosti $[0,\infty)$



← se odločimo da na obeh straneh vzamemo $\frac{\alpha}{2}$

$$P(\delta_- < Q < \delta_+) \geq 1-\alpha$$

$$\delta_+ = F^{-1}_{\chi^2_{n-1}}\left(1-\frac{\alpha}{2}\right)$$

$$\delta_- = F^{-1}_{\chi^2_{n-1}}\left(\frac{\alpha}{2}\right)$$

$$P(\delta < Q < \delta_+) = 1-\alpha$$

$$P\left(\underbrace{\frac{(n-1)\hat{\sigma}^2}{\delta} < \sigma < \frac{(n-1)\hat{\sigma}^2}{\delta}}_{\text{iz}}\right) = 1-\alpha$$

Definiramo kvantilno funkcijo $\tilde{F}^{-1}$ kot

$$\tilde{F}^{-1} : (0,1) \to \mathbb{R}$$

$$\alpha \mapsto \min \{x \in \mathbb{R} \mid F(x) \geq \alpha\}$$

Pokaži da je $\tilde{F}^{-1}(\alpha)$ kvantil reda α

$$y = \tilde{F}^{-1}(\alpha) \qquad F(y) \geq \alpha$$

Vzemimo $x < y \Rightarrow F(x) < \alpha$

$$\lim_{x \to y} F(x) < \alpha \Rightarrow F(y) \leq \alpha$$

Trditev: Naj bo $U \sim U(0,1)$. definiramo $X = \tilde{F}^{-1}(U)$ potem je X porazdeljen z F

lema: $\tilde{F}^{-1}(\alpha) \leq x \iff \alpha \leq F(x)$

$(\Leftarrow)$ Naj bo $\alpha \leq F(x)$

$$\tilde{F}^{-1}(\alpha) = \min\{y \in \mathbb{R} \ F(y) \geq \alpha\}$$
$$\Rightarrow F(\tilde{F}^{-1}(\alpha)) \leq F(x) \leadsto \tilde{F}^{-1}(\alpha) \leq x$$

ker je F naraščajoča

$(\Rightarrow)$

$$\tilde{F}^{-1}(\alpha) \leq x$$
$$y \quad F(y) \geq \alpha \Rightarrow F(x) \geq \alpha$$
$$\alpha \leq F(y) \leq F(x)$$

Dokaz trditve 127

$X \sim F$

$P(X \leq x) = F(x)$

$$P(\tilde{F}^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$$

Točkasto ocenjevanje

Imamo vzorec X (vektor) in s porazdelitvijo povezano količino ϑ

količino ϑ ocenimo s cenilko $\hat{\vartheta} = f(X)$

$X_1, \dots, X_n$ NEP

$X_n \sim U(0, a)$ $a > 0$

oceni a

vemo: $E(X_n) = \frac{a}{2} = \mu$

$\bar{x} = \hat{\mu}$

$\hat{a} = 2\bar{x}$

$MSE(\hat{a}) = ?$ $MSE = Var + Bias^2$

$$Bias(\hat{a}) = E(2\bar{x}) - a = 2 \cdot \frac{a}{2} - a = 0$$

$$var(\hat{a}) = var\left(\frac{2\sum x_i}{n}\right) = \sum var(x_i) = 4 \cdot \frac{a^2}{12} \cdot n = \frac{a^2 n}{3n^2}$$

$$MSE = \frac{a^2}{3n} \xrightarrow[n \to \infty]{} 0 \Rightarrow \hat{a} \text{ je dosledna}$$

17. 4.

$X_i \sim N(\mu, \sigma^2)$. Pri katerem c bo cenilka najboljša

$$MSE(c\hat{\sigma}^2) = c^2\left(\frac{(n-1)^2}{n^3}\mu^4 + \frac{(n-2n+3)(n-1)}{n^3}\sigma^4\right) - 2c\sigma^4\frac{n-1}{n} + \sigma^4$$

$$= \left(c^2\frac{n^2-1}{n^2} - c\cdot\frac{2(n-1)}{n} + 1\right)\sigma^4$$

$$\frac{\partial}{\partial c}(\ldots) = \left(c\,\frac{2(n^2-1)}{n^2} - \frac{2(n-1)}{n}\right)\sigma^4 = 0$$

$$\rightsquigarrow \quad c = \frac{n^2}{2(n^2-1)}\cdot\frac{2(n-1)}{n} = \frac{n}{n+1}$$

X, Y slučajna vektorja ~~te~~ dimenzije
Poznamo $\mathrm{var}(X) = \Sigma_{11}$ $\mathrm{var}(Y) = \Sigma_{22}$
$\mathrm{cov}(X, Y) = \Sigma_{12}$

$$\mathrm{Var}(X+Y) = \mathrm{Cov}(X+Y, X+Y) =$$

$$= \mathrm{cov}\left([I_n, I_n]\begin{bmatrix} X \\ Y \end{bmatrix}, [I_n, I_n]\begin{bmatrix} X \\ Y \end{bmatrix}\right) =$$

$$[I_n, I_n]\,\mathrm{cov}\left(\begin{bmatrix} X \\ Y \end{bmatrix}, \begin{bmatrix} X \\ Y \end{bmatrix}\right)[I_n, I_n]^T =$$

$\mathrm{cov}(X)$ . . .

nek

$X \sim QX$ za $\forall Q$ ortonormirana

$$Q^T Q = I$$

$$\mathrm{var}(X) = \begin{bmatrix} \mathrm{cov}(x_1, x_1), \dots & \\ & \ddots \\ \dots & \end{bmatrix}$$

$$\mathrm{var}(X) = \mathrm{var}(QX) = Q\,\mathrm{var}(X)\,Q^T$$

$$\Rightarrow Q\,\mathrm{var}(X) = \mathrm{var}(X)\,Q$$

A simetrična $\Rightarrow \exists Q$ $A = QDQ^T$ D diag

$\Rightarrow \mathrm{var}(X)$ diagonal

$E(X) = 1_n \cdot a \qquad var(X) = I_n + b 1_n \cdot 1_n^T$

$E(X^T(I + \frac{1}{n} 1_n \cdot 1_n^T) X)$

$y = X^T(I + \frac{1}{n} 1_n \cdot 1_n^T) X$

$E(y) = E(sl(y))$, ker je y skalar

$sl(AB) = sl(BA)$

$\Rightarrow sl(y) = sl((I + \frac{1}{n} 1_n \cdot 1_n^T) X X^T)$

$E(sl(y)) = sl(I + \frac{1}{n} 1_n \cdot 1_n^T \underbrace{E(XX^T)}_{var(X) + E(X)E(X)^T}) =$

$\Rightarrow E(y) = sl((I + \frac{1}{n} 1_n \cdot 1_n^T)(I_n + b \cdot 1_n \cdot 1_n^T + a^2 1_n \cdot 1_n^T)) =$

$= sl(I_n + 2(\frac{1}{n} + b + a^2) 1_n 1_n^T + \frac{1}{n}(b + a^2) n 1_n 1_n^T) =$

$= 1 + 2(\frac{1}{n} + b + a^2) + b + a^2 = \underline{\underline{1 + \frac{2}{n} + 3b + 3a^2}}$

Manjkala sem enkrat prej

$X_1, \dots X_n$ MEP

$f_X(x, \vartheta, \lambda) = \lambda e^{-\lambda(x-\vartheta)} \cdot \mathbb{1}_{x \geq \vartheta}$

a) Določi cenilki po metodi momentov ϑ, λ

$$\mathbb{E}(X) = \int_{\mathbb{R}} x \, f_X(x, \vartheta, \lambda)\, dx$$

$\mathbb{R} = x \geq \vartheta$

Opazimo $X \stackrel{d}{=} Y + \vartheta \quad Y \sim Exp(\lambda)$

$$\Rightarrow E(X) = \frac{1}{\lambda} + \vartheta$$

$$var(X) = \frac{1}{\lambda^2}$$

$$\mathbb{E}(X^2) = var(X) + \mathbb{E}(X)^2 = \frac{2}{\lambda^2} + 2\frac{\vartheta}{\lambda} + \vartheta^2$$

$$\vartheta = E(X) - \frac{1}{\lambda}$$

$$\mathbb{E}(X^2) = \frac{1}{\lambda^2} + \mathbb{E}(X^2)$$

$$\rightsquigarrow \lambda = \frac{1}{\sqrt{E(X^2) - E^2(X)}}$$

$$\vartheta = E(X) - \sqrt{E(X^2) - E^2(X)}$$

$$\lambda = \frac{1}{\sqrt{\overline{X^2} - \overline{X}^2}} \qquad \vartheta = \overline{X} - \sqrt{\overline{X^2} - \overline{X}^2}$$

b) Cenilki po metodi največjega verjetja

$$L(\lambda, \vartheta | x_1, \dots x_n) = \prod_{i=1}^{n} f_X(x_i)$$

$$= \lambda^n e^{-\lambda(\sum x_i - n\vartheta)} \mathbb{1}_{x_1 \geq \vartheta} \cdots \mathbb{1}_{x_n \geq \vartheta}$$

log

$$l = \ln L = \ln \lambda^n - \lambda(\sum x_i - n\vartheta) + \sum_{i=1}^{n} \ln(\mathbb{1}_{x_i \geq \vartheta})$$

$$\frac{\partial}{\partial \lambda} l = n \cdot \frac{1}{\lambda} - \sum x_i + n\vartheta + \sum_{i=1}^{n} \ln(\mathbb{1}_{x_i \geq \vartheta})$$

L hočemo maksimizirat $\Rightarrow$ hočemo l maksimizirat

$\vartheta: \; x_i \geq \vartheta \quad \forall i$

zaradi člena $+n\vartheta$ želimo ϑ čim večjo

$$\Rightarrow \hat{\vartheta} = \min(x_i) = X_{(1)}$$

oznaka da je najmanjši

$$\frac{n}{\lambda} = \sum x_i - n X_{(1)} + \underbrace{n \ln(1)}_{0}$$

$$\hat{\lambda} = \frac{n}{\sum x_i - n X_{(1)}} = \underline{(\overline{X} - X_{(1)})^{-1}}$$

$X_1, \ldots, X_n$ NEP

$$f_X(x,\vartheta) = e^{\vartheta - x} \mathbb{1}_{x>\vartheta} \quad ; \vartheta > 0 \text{ neparam}$$

Določi ϑ MNV

$$L = \prod_{x_i} f_X(x_i,\vartheta) = e^{-\sum x_i + n\vartheta} \mathbb{1}_{x>\vartheta} = L(\vartheta \mid x_1 \ldots x_n)$$

$$\ln L = \ell = n\vartheta - \sum x_i + \ln \mathbb{1}_{x>\vartheta}$$

maksimiziramo: $n\vartheta + \ln \mathbb{1}_{x>\vartheta}$

Poljubno malo pa lahko vzamemo

Rabimo $\min(x_i) - \varepsilon$

Ampak noben ε ni najboljši

$\Rightarrow$ cenilka po MNV ne obstaja

v praksi je isto če vzamemo $\leq$ tak

je $\min(x_i)$ najboljša cenilka

Fisherjeva informacija

Pri modelih $X=(X_1 \ldots X_n)$ in z neznanim parametrom $\vartheta \in \mathbb{R}$, je dana z

$$FI(\vartheta) = \mathbb{E}\left(\left(\frac{\partial l(\vartheta|X)}{\partial \vartheta}\right)^2\right) = -\mathbb{E}\left(\frac{\partial^2 l(\vartheta|X)}{\partial \vartheta}\right)$$

Velja Rao-Cramerjeva ocena

$$SE^2 = var(\hat{\vartheta}) \geq \frac{1}{FI(\vartheta)}$$

za nepristransko cenilko $\hat{\vartheta}$

Če je model dovolj regularen za cenilko $\hat{\vartheta}$ po MNV velja $\hat{\vartheta} \to \vartheta$ $\quad \mathbb{E}(\hat{\vartheta}) \to \vartheta$

$$SE^2 \sim \mathbb{E}\left((\hat{\vartheta} - \vartheta)^2\right) \sim \frac{1}{F(\vartheta)}$$

Če so $X_1, \ldots, X_n$ NEP je $FI(\vartheta) = n FI_1(\vartheta)$

$$FI_1(\vartheta) = \mathbb{E}\left(\frac{\partial l_1(\vartheta|X_1)}{\partial \vartheta}\right)^2$$

$X_1, \dots, x_n \sim Exp\left(\frac{1}{\alpha}\right)$ NEP

Določi Fisherjevo transformacijo za α

$f_X(x) \frac{1}{\alpha} e^{-\frac{1}{\alpha} x}$

$\cancel{l = \ln \frac{1}{\alpha} + \sum \frac{1}{\alpha} x_i}$

$$l_1 = \ln \frac{1}{\alpha} - \frac{1}{\alpha} x_1 = -\ln \alpha - \alpha^{-1} x_1$$

$$\frac{\partial l_1}{\partial \alpha} = -\frac{1}{\alpha} + \frac{1}{\alpha^2} x_1$$

$$\frac{\partial l_1}{\partial \alpha} = \frac{1}{\alpha^2} - 2 \frac{1}{\alpha^3} x_1$$

$$F_I(\alpha) = n \mathbb{E}\left(2 \frac{1}{\alpha^3} x_1 - \frac{1}{\alpha^2}\right) =$$

$$= 2n \frac{1}{\alpha^3} \underbrace{\mathbb{E}(X_1)}_{=\alpha} - \frac{n}{\alpha^2} = \frac{n}{\alpha^2}$$

od zadnjič vemo: $\hat{\alpha} = \bar{X}$ sledi

$$SE^2 \geq \frac{1}{FI(\alpha)} = \frac{\alpha^2}{n}$$

□

Ocena F. inf. je $FI(\hat{\alpha})$.

Nadaljevanje yapa od prej:

Če je $\Theta \in \mathbb{R}^d$ in imamo karakteristiko $\xi = h(\vartheta)$ po MNV in ker $\hat{\xi} = h(\hat{\vartheta})$ za ϑ po MNV.

Velja $$FI(\vartheta) = \left[E\left(\frac{\partial l(\vartheta | x)}{\partial \vartheta_i} \frac{\partial l(\vartheta | x)}{\partial \vartheta_j} \right) \right]_{i,j=1}^{d} =$$
$$= -\left[E\left(\frac{\partial l^2(\vartheta | x)}{\partial \vartheta \partial \vartheta_j} \right) \right]_{i,j=1}^{d}$$

Če je $\hat{\xi}$ nepristranska je
$$SE^2 = var(\hat{\xi}) \geq \nabla h(\vartheta)^T FI^{-1}(\vartheta) \nabla h(\vartheta)$$

Preostala isto kot $\Gamma \xi$

Od zadnjič $X_1 \ldots X_n$ NEP

$X_1 \sim \begin{pmatrix} -1 & 0 & 1 & 2 \\ 1-2a-b & a & b & a \end{pmatrix}$

$\hat{a} = \frac{1}{4}$

$\hat{b} = \frac{3}{10}$

ocenjujemo $a^2 \cdot b = h(a,b) = \xi$

a) oceni ξ po MNV

b) poišči asimptotič $SE(\xi)$

a) $\hat{\xi} = h(\hat{a}, \hat{b}) = \frac{3}{160}$

b) $\nabla h(a,b)^T FI^{-1}(a\,b) \nabla h(a,b)$

$\nabla h(a,b)^T = \begin{bmatrix} 2ab \\ a^2 \end{bmatrix}$

$FI(a,b) = ?$

$$\frac{l_1}{\partial a} = \begin{cases} \frac{-2}{1-2a-b} \\ \frac{1}{a} & ;\ 0 \\ 0 & ;\ 1 \\ \frac{1}{a} & ;\ x=2 \end{cases}$$

$$\frac{l_1}{\partial b} = \begin{cases} \frac{-1}{1-2a-b} & -1 \\ 0 & 0 \\ \frac{1}{b} & 1 \\ 0 & 2 \end{cases}$$

$$E\left(\frac{\partial l_1(a,b|x)}{\partial a}\right)^2 = \sum_{i=-1}^{2} (\ldots)^2 P(X=i)$$

$$= \frac{4}{(1-2a-b)^2}(1-2a-b) + 2\frac{1}{a^2}\cdot a + 0 =$$

$$= \frac{4}{1-2a-b} + \frac{2}{a^2} = \frac{2-2b}{a(1-2a-b)}$$

$E\left(\frac{}{\partial a}\ \frac{}{\partial b}\right) = \frac{2}{1-2a-b}$

$E\left(\left(\frac{}{\partial b}\right)^2\right) = \frac{1-2a}{b(1-2a-b)}$

$FI_n(a,b) =$

$$Fl_1(a,b) = \frac{1}{1-2a-b}\begin{bmatrix} \frac{2-2b}{a} & 2 \\ 2 & \frac{1-2a}{b} \end{bmatrix}$$

$$\Rightarrow Fl_1^{-1} = \frac{?}{2}\begin{pmatrix} a(1-2a) & -2ab \\ -2ab & b(2-b) \end{pmatrix}$$

$Fl^{-1} = \frac{1}{n} Fl_1^{-1}$

b) $\nabla h^T FI^{-1} \nabla h = \left(\frac{a^3 b}{n}(a - 3ab + 2b)\right)^{\frac{1}{2}}$

Standardna p-razsežna normalna porazdelitev je porazdelitev vektorja

$$Z = \begin{bmatrix} Z_1 \\ \vdots \\ Z_n \end{bmatrix}$$ kjer so Z_i NEP.

Splošna n-razsežna normalna porazdelitev je vsaka porazdelitev slučajnega vektorja

$$AZ + \mu$$

Naloga

$$\begin{bmatrix}X\\Y\end{bmatrix} \sim N\left(0, \begin{bmatrix}8&2\\2&5\end{bmatrix}\right)$$

Določi $P(X>0, Y>0)$

$\exists A \qquad \Sigma = AA^T$

$AZ \sim N(0, I)$

$$a_{jj} = \sqrt{\Sigma_{jj} - \sum_{k}^{j-1} a_{jk}^2}$$

$$i>j: \frac{\Sigma_{ij} - \sum_{k=1}^{j-1} a_{ik} a_{jk}}{a_{jj}}$$

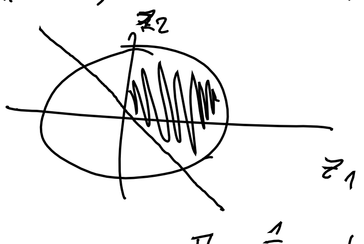
$$\begin{bmatrix}\sqrt{8} & \frac{1}{\sqrt{2}}\\ 0 & \sqrt{5-\frac{1}{2}}\end{bmatrix} = \begin{bmatrix}2\sqrt{2} & \frac{\sqrt{2}}{2}\\ 0 & \frac{3\sqrt{2}}{2}\end{bmatrix} = A^T$$

$$V = AZ = \begin{bmatrix}X\\Y\end{bmatrix}$$

$P(X>0, Y>0$

$$P\left(2\sqrt{2} Z_1 > 0, \frac{\sqrt{2}}{2} Z_1 + \frac{3\sqrt{2}}{2} Z_2 > 0\right) =$$

$$= P\left(Z_1 > 0, Z_2 > -\frac{1}{3} Z_1\right)$$



$$P = \text{razmerje} = \frac{\frac{\pi}{2} + \frac{1}{3}\arctan\frac{1}{3}}{\pi} =$$

$$= \frac{1}{2} + \frac{1}{3\pi}\arctan\frac{1}{3}$$

Naj bo $Z \sim N(\vec{0}, I_n)$. Potem je $\|Z\| \sim \chi^2(n) = \Gamma(\frac{n}{2}, \frac{1}{2})$

studentova porazdelitev.

$$T = \frac{Z}{\sqrt{\frac{1}{p} K}} \qquad Z \sim N(0,1) \qquad K \sim \Gamma(\frac{p}{2}, \frac{1}{2}) = \chi^2(p)$$

Z, K neodvisni

$T \sim student(p)$

$$\frac{\bar{X} - \mu}{\sigma} \sqrt{n} \sim N(0,1)$$

$$\hat{\sigma}_+ = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2}$$

$$\frac{\bar{X} - \mu}{\hat{\sigma}_+} \sqrt{n} = \frac{\overbrace{\frac{\bar{X} - \mu}{\sigma} \sqrt{n}}^{\sim N(0,1)}}{\sqrt{\frac{\sigma_+^2}{\sigma^2}}} =$$

$$\frac{\sigma_+}{\sigma} = \sqrt{\frac{1}{\sigma^2} \frac{1}{n-1} \sum (X_i - \bar{X})^2} \qquad = \|HX\|^2 \sim \chi^2(n-1)$$

Neodvisnost?

Povdj je pokazati, da sta neodvisna $\bar{X}$ in HX

Ker sta $\bar{X}$ in HX normalne slučajne veltorja neodvisnost sledi iz nekoreliranosti

$$\mathrm{cov}(\bar{X}, HX) = \mathrm{cov}\left(\frac{1}{n} 1_n^T X, HX\right) =$$

$$= \frac{1}{n} 1^T \mathrm{Cov}(X, X) H^T \overset{= H}{=} \frac{1}{n} 1^T \sigma^2 H =$$

$$= \frac{1}{n} 1_n^T \sigma^2 \left(I_n - \frac{1}{n} 1_n 1_n^T\right) = 0$$

Naj bodo $X_1, \dots, X_n$ NEP $N(\mu, \sigma^2)$
Skonstruiraj interval zaupanja za stopnje $1-\alpha$ za:
a) μ, če poznamo σ?

$\hat{\mu} = \overline{X}$ $\qquad c_\alpha = ?$

$$\mathbb{P}(\overline{X} - c_\alpha < \mu < \overline{X} + c_\alpha) \geq 1-\alpha$$

$$\mathbb{P}(\mu - c_\alpha \leq \overline{X} \leq \mu + c_\alpha) =$$

$$\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$Y = \sqrt{n}\frac{\overline{X}-\mu}{\sigma} \sim N(0,1)$$

$\phi(-x) = 1 - \phi x$

$$= \mathbb{P}(-\frac{\sqrt{n}}{\sigma}c_\alpha \leq Y \leq \frac{\sqrt{n}}{\sigma}c_\alpha) =$$

$$= \Phi(\frac{\sqrt{n}}{\sigma}c_\alpha) - \Phi(-\frac{\sqrt{n}}{\sigma}c_\alpha) =$$

$$= 2\Phi(\frac{\sqrt{n}}{\sigma}c_\alpha) - 1 \geq 1-\alpha$$

$$\Rightarrow \Phi(\frac{\sqrt{n}}{\sigma}c_\alpha) \geq 1 - \frac{\alpha}{2}$$

$$c_\alpha \geq \frac{\sigma}{\sqrt{n}}\Phi^{-1}(1-\frac{\alpha}{2})$$

b) stopnja zaupanja za μ, če σ ni poznan

$$\hat{\sigma}_+ = \sqrt{\frac{1}{n}\sum (x_i - \bar{x})^2}$$

$$\mathbb{P}(\bar{X} - c_\alpha < \mu < \bar{X} + c_\alpha) \leq 1 - \alpha$$

$$T = \frac{\bar{X} - \mu}{\hat{\sigma}_+}\sqrt{n} \sim \text{student}(n-1)$$

$$\mathbb{P}(\mu - c_\alpha < \bar{X} < \mu + c_\alpha) =$$

$$= \mathbb{P}\left(-\frac{c_\alpha}{\hat{\sigma}_+}\sqrt{n} < T < \frac{c_\alpha}{\hat{\sigma}_T}\sqrt{n}\right) = 1 - \alpha$$

$$F\left(+\frac{c_\alpha}{\hat{\sigma}_+}\sqrt{n}\right) - F\left(-\frac{c_\alpha}{\hat{\sigma}_+}\sqrt{n}\right) = 1 - \alpha$$

$$2F\left(c_\alpha \frac{\sqrt{n}}{\hat{\sigma}_+}\right) - 1 = 1 - \alpha$$

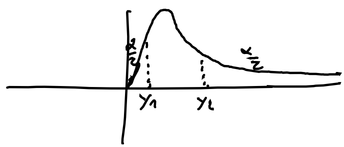
$$\Rightarrow \quad c_\alpha = \frac{\hat{\sigma}_+}{\sqrt{n}} F^{-1}\left(1 - \frac{\alpha}{2}\right)$$

c) za σ, če μ ni poznan

$$\cancel{[\sigma_+ - c_\alpha, \sigma_+ + c_\alpha]}$$

$$\frac{\hat{\sigma}_+^2}{\sigma^2} = \frac{1}{\sigma^2}\frac{1}{n-1}\sum (x_i - \overline{x})^2 \sim \chi^2(n-1)$$

$$\mathbb{P}\left(y_1 \leq \frac{\hat{\sigma}_+^2}{\sigma^2} \leq y_2 \right) = 1-\alpha$$



$$P\left(F^{-1}\left(\frac{\alpha}{2}\right) \leq \frac{\hat{\sigma}_+^2}{\sigma^2} \leq F^{-1}\left(1-\frac{\alpha}{2}\right)\right) =$$

$$= P\left(\frac{1}{F(1-\frac{\alpha}{2})}\hat{\sigma}_+^2 \leq \sigma^2 \leq \frac{\hat{\sigma}_+^2}{F^{-1}(\frac{\alpha}{2})}\right)$$

interval zaupanja: $\left[\sqrt{\frac{\hat{\sigma}_+^2}{F(1-\frac{\alpha}{2})}}, \sqrt{\frac{\hat{\sigma}_+^2}{F^{-1}(\frac{\alpha}{2})}}\right.$

Naj bo $\hat{\vartheta}$ cenilka pa MNV po dovolj lepem modelu

Pri dovolj velikih n je $\hat{\vartheta}$ porazdeljena približno

$$N\left(\vartheta, \frac{1}{FI(\vartheta)}\right)$$

To nam da asimptotični interval zaupanja $1-\alpha$ dolžine

$$\left[\hat{\vartheta} - \frac{F^{-1}_{N(0,1)}\left(1-\frac{\alpha}{2}\right)}{\sqrt{FI(\hat{\vartheta})}}, \hat{\vartheta} + \dots\right.$$

$X_1, \dots X_n \qquad X_i \sim N(\mu, \sigma^2)$.

IZ za σ

$$\left[\hat{\sigma}_+^2 \sqrt{\frac{n-1}{F^{-1}_{\chi^2(n-1)}(1-\frac{\alpha}{2})}} \, , \, \sigma_+^2 \sqrt{\frac{n-1}{F^{-1}_{\chi^2(n-1)}(\frac{\alpha}{2})}} \right]$$

$$\sigma_+ = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

asimptotski ?

$L(\mu, \sigma)$

$$L_1 = (\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$l(\mu, \sigma) = \sum_{i=1}^{n} \ln\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{(x-\mu)^2}{2\sigma^2}$$

Izkaže se $\hat{\mu} = \bar{X}$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_i^n (x_i - \bar{x})^2$$

$$\frac{\partial^2}{\partial \mu^2} l_1 = -\frac{1}{\sigma^2}$$

$$\frac{\partial^2}{\partial \mu \sigma} l_1 = -\frac{2(X_1-\mu)}{\sigma^3}$$

$$\frac{\partial^2}{\partial \sigma^2} l_1 = \frac{1}{\sigma^2} - \frac{3(X_1-\mu)^2}{\sigma^4}$$

$$FI_1(\mu, \sigma) = -E \begin{bmatrix} -\frac{1}{\sigma^2} & -\frac{2(X_1-\mu)}{\sigma^3} \\ -\frac{2(X_1-\mu)}{\sigma^3} & \frac{1}{\sigma^2} - \frac{3(X_1-\mu)^2}{\sigma^4} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sigma^2} & 0 \\ 0 & -\frac{1}{\sigma^2} + \frac{3}{\sigma^2} \end{bmatrix}$$

$$\sigma^2 = h(\vartheta) = pr_2(\vartheta) \qquad \vartheta = \begin{bmatrix} \mu \\ \sigma \end{bmatrix}$$

$$\nabla h^T FI^{-1} \nabla h$$

$$[0, 1] \begin{bmatrix} \cdot & \\ & \cdot \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{\sigma^2}{2n} \sim var(\hat{\sigma}^2)$$

$$\hat{\sigma}^2 - \frac{F^{-1}_{N(0,1)}(1-\frac{\alpha}{2})}{\sqrt{\frac{2n}{\hat{\sigma}}}} \, , \, \hat{\sigma}^2 + \frac{F^{-1}(1-\frac{\alpha}{2})}{\sqrt{\frac{\hat{\sigma}^2}{2n}}^{-1}}$$

Domneva H_0 in H

če je model parametriziran s $\vartheta \in \Theta$

lahko H_0 določimo z $\vartheta \in \Theta_0$

Alternativno domeno lahko določimo

kar z $\Theta_1 = \Theta - \Theta_0$

Pri stopnji tveganja α skonstruiramo

preizkus, da velja

$P_0(H_0 \text{ zavr.}) = P(H_0 \text{ zavrnemo} \mid H_0 \text{ drži}) \leq \alpha$

Moč preizkusa

$P(H_0 \text{ zavrnemo} \mid H_1 \text{ drži})$

Odločitev o zavrnitvi H_0 sprejmemo, če preizkusna statistika pade v kritično območje K_α

	H_0 drži	H_1 drži
H_0 zavrnemo	Napaka tipa 1 $P \leq \alpha$	✓ $P(\ldots) = 1-\beta$ ← moč testa
H_0 ne zavrnemo	✓	Napaka tipa 2 $P(\ldots) = \beta$

Opazimo $X_1, \dots X_n$ NEP $X_i \sim N(\mu, \sigma^2)$

Vemo $\sigma = 5$

Preizkusamo $H_0: \mu = 100$

a) $H_1: \mu < 100$

Preizkus stopnje tveganja α

$$\bar{X} = \frac{1}{n}(X_1 + \dots + X_n)$$

Zavrnemo če: $\bar{X} < k_\alpha$

$$P(\bar{X} < k_\alpha \mid \mu < 100) =$$

iščemo ~~najmanjši~~ največji možni k_α, da je moč testa največja

$$P_0(\bar{X} < k_\alpha) \leq \alpha$$

$$X_1 \sim N(100, 5^2)$$

$$\bar{X} \sim N\left(100, \frac{5^2}{n}\right)$$

$$var\left(\frac{1}{n}\bar{X}\right) = \frac{1}{n^2}\sum var(X_i) = \frac{1}{n} var(X_1)$$

$$P\left(\frac{\bar{X} - 100}{5}\sqrt{n} < \frac{k_\alpha - 100}{5}\sqrt{n}\right) \leq \alpha$$

$$\Phi\left(\frac{k_\alpha - 100}{5}\sqrt{n}\right) \leq \alpha$$

$$k_\alpha = \frac{5}{\sqrt{n}}\Phi^{-1}(\alpha) + 100$$

b) $H_1: \mu \neq 100$

$\mu_0 - c \quad \mu_0 \quad \mu_0 + c$

$$P_0(\bar{X} - c \leq \mu \leq \bar{X} + c) = P(100 - c \leq \bar{X} \leq 100 + c) =$$

$$= P\left(-\frac{c\sqrt{n}}{5} \leq \frac{\bar{X} - 100}{5}\sqrt{n} \leq \frac{c\sqrt{n}}{5}\right) =$$

$$1 - \left(\Phi\left(\frac{c\sqrt{n}}{5}\right) - \Phi\left(-\frac{c\sqrt{n}}{5}\right)\right) = \alpha$$

$$1 - \alpha = -1 + 2\Phi\left(\frac{c\sqrt{n}}{5}\right)$$

$$\frac{c\sqrt{n}}{5} = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$$

$$c = \frac{5}{\sqrt{n}}\Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$$

$X_1 \ldots X_5$ NEP $\sim N(1, \sigma^2)$

Z verjetnostjo vsaj 0.95 se želi [illegible] kaj je σ^2 [illegible] [illegible] 0.256

$$\hat{\sigma}^2 = \frac{1}{5} \sum^{n} (X_i - 1)^2$$

$$T_\sigma = \frac{\hat{\sigma}^2 \cdot 5}{\sigma^2} \sim \chi_2(5) \quad \text{če je} \quad \mathrm{var}(X_i) = \sigma_0^2$$

kar je pri $\sigma^2 = \sigma_0^2$ odločitev najbolj tvegana

Investiramo, če $T < c$

$\mathbb{P}_{\sigma^2}(T < c)$ je padajoča kot funkcija σ^2

$\Rightarrow 1 - P_\sigma(T < c)$ naraščajoča v σ^2 torej je najbolj kritična odločitev pri σ_0^2

$$\mathbb{P}_{\sigma_0^2}(T_{\sigma_0^2} < c) = \alpha$$

$$c = F^{-1}_{\chi_2(5)}(\alpha)$$

Investiramo če $\dfrac{\hat{\sigma}^2 \, 5}{\sigma_0} < F^{-1}_{\chi^2}(\alpha)$

Preizkus na podlagi razmerja verjetij

Imejmo: H_0 s Θ_0

H_1 s $\Theta_1 = \Theta - \Theta_0$

Preizkus na podlagi raz. verjetij temelji na statistiki

$$\Lambda = \frac{\sup\limits_{\vartheta \in \Theta} (L(\vartheta | X))}{\sup\limits_{\vartheta \in \Theta_0} (L(\vartheta | X))}$$

oz

$$\Lambda' = \frac{\sup\limits_{\vartheta \in \Theta_1} L(\vartheta | X)}{\sup\limits_{\vartheta \in \Theta_0} L(\vartheta | X)}$$

Wilksov izrek

$$2 \ln \Lambda \xrightarrow[n \to \infty]{d} \chi^2(\dim \Theta - \dim(\Theta_0))$$

X ... št klicev v kopru (dnevno)
Y ... št licev v celju

$\bar{X} = 20 \quad \bar{y} = 22$ št klicev $\sim Po(\lambda_x)$
$\sim Po(\lambda_y)$

$H_0: \lambda_x = \lambda_y$

$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$

$L(\lambda_x, \lambda_y | x,y) = \prod_{i=1}^{n} L(\lambda_{x_i}, \lambda_{x_2} | x_i, y_i)$

$\Theta = [0, \infty)^2$
$\Theta_0 = [0,\infty)$

$l(\lambda_x, \lambda_y | xy) = -n\lambda_x - n\lambda_y + \sum x_i \ln(\lambda_x) - \ln(x_i!) + y_i \ln(\lambda_y)$
$- \ln(y_i!)$

a) $l = -2n\lambda + \sum (x_i + y_i)\ln\lambda$

$\frac{\partial l}{\partial \lambda} = 2n + \sum \frac{x_i + y_i}{\lambda} = 0$

$\Rightarrow \lambda = \frac{\sum x_i + y_i}{2n} = \frac{1}{2}(\bar{x} + \bar{y})$

b) $\frac{\partial l}{\partial \lambda_x}, \frac{\partial l}{\partial \lambda_y} \dots \lambda_x = \bar{x} \quad \lambda_y = \bar{y}$

$\Lambda = \frac{\prod \frac{\bar{x}^{x_i}}{x_i!} e^{-\bar{x}} \frac{\bar{y}^{y_i}}{y_i!} e^{-\bar{y}}}{\prod \frac{(\frac{1}{2}(\bar{x}+\bar{y}))^{x_i}}{x_i! y_i!} e^{-(\bar{x}+\bar{y})}} =$

$= \prod_{i=1}^{n} \left(2\frac{\bar{x}\bar{y}}{\bar{x}+\bar{y}}\right)^{x_i} = \left(2\frac{\bar{X}\bar{Y}}{\bar{X}+\bar{Y}}\right)^{n\bar{x}+n\bar{y}}$

Uporabimo Wilksov izrek

$2\ln(\Lambda) \xrightarrow[n\to\infty]{d} \chi^2(1)$

$\mathbb{P}\left(2n(\bar{x}+\bar{y})\ln\left(\frac{\bar{X}\bar{Y}}{\bar{X}+\bar{Y}}\right) > d\right) = \alpha$

$F^{-1}_{\chi^2(1)}(1-\alpha) = d$

~~sprejmemo~~ zavrnemo za $2\ln\Lambda > d$

Linearna regresija

Model $\quad y = X\beta + \varepsilon$

y slučajen, opažen

cilj: ocenimo β in varianco ε

$$\boxed{\text{NNLC za } \beta \text{ je} \quad \hat{\beta} = (X^T X)^{-1} X^T y}$$

Na realni osi imamo $\beta_1, \beta_2, \beta_3$

Izmerimo $\beta_1, \beta_2, \beta_3, \beta_1 - \beta_2, \beta_2 - \beta_3, \beta_1 - \beta_3$

$$y_1 = \beta_1 + \varepsilon_1$$
$$y_2 = \beta_2 + \varepsilon_2$$
$$y_3 = \beta_3 + \varepsilon_3$$
$$y_4 = \beta_1 - \beta_2 + \varepsilon_4$$
$$y_5 = \beta_2 - \beta_3 + \varepsilon_5$$
$$y_6 = \beta_1 - \beta_3 + \varepsilon_6$$

$$\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \\ 1 & -1 & \\ & 1 & -1 \\ 1 & & -1 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T y$$

Velja:

$$\operatorname{var} \hat{\beta} = \sigma^2 (X^T X)^{-1}$$

$$\hat{\sigma}^2 = \frac{\| Y - X\hat{\beta} \|^2}{n-m}$$

nepristranske cenilke za σ^2

Če je model gaussov ($\varepsilon \sim N(0, \sigma^2 I)$) je

$$\frac{\hat{\beta} - \beta}{\sigma} \sim N(0, (X^T X)^{-1})$$

za $\frac{\hat{\sigma}^2 (n-m)}{\hat{\sigma}} \sim \chi^2(n-m)$ ← pojasnjevalne ← koliko točk napovedujemo

št podatkov